# STACKS, DEFORMATION THEORY: LEMMA-CANONICAL-CLASS-ALGEBRA

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 5134-5159 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5161-5244, the complete author proof (82 body lines).

*Editorial note.* Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 8 of queue_defos.json (pb=3263, stmt 5134-5159, proof 5161-5244).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

**Lemma 0.1.** *Let $R' \to R$ be a surjection of rings whose kernel is an ideal $I$ of square zero. For every $K \in D^-(R)$ there is a canonical map*

$$\omega(K) : K \longrightarrow K \otimes^{\mathbf{L}}_R I[2]$$

*in $D(R)$ with the following properties*

(1) *$\omega(K) = 0$ if and only if there exists $K' \in D(R')$ with $K' \otimes^{\mathbf{L}}_{R'} R = K$,*
(2) *given $K \to L$ in $D^-(R)$ the diagram*

$$\begin{array}{ccc} K & \xrightarrow[\omega(K)]{} & K \otimes^{\mathbf{L}}_R I[2] \\ \downarrow & & \downarrow \\ L & \xrightarrow{\omega(L)} & L \otimes^{\mathbf{L}}_R I[2] \end{array}$$

*commutes, and*
(3) *formation of $\omega(K)$ is compatible with ring maps $R' \to S'$ (see proof for a precise statement).*

*Proof.* Choose a bounded above complex $K^\bullet$ of free $R$-modules representing $K$. Then we can choose free $R'$-modules $(K')^n$ lifting $K^n$. We can choose $R'$-module maps $(d')^n_K : (K')^n \to (K')^{n+1}$ lifting the differentials $d^n_K : K^n \to K^{n+1}$ of $K^\bullet$. Although the compositions

$$(d')^{n+1}_K \circ (d')^n_K : (K')^n \to (K')^{n+2}$$

may not be zero, they do factor as

$$(K')^n \to K^n \xrightarrow{\omega^n_K} K^{n+2} \otimes_R I = I(K')^{n+2} \to (K')^{n+2}$$

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because $d^{n+1} \circ d^n = 0$. A calculation shows that $\omega_K^n$ defines a map of complexes. This map of complexes defines $\omega(K)$.

Let us prove this construction is compatible with a map of complexes $\alpha^\bullet : K^\bullet \to L^\bullet$ of bounded above free $R$-modules and given choices of lifts $(K')^n, (L')^n, (d')^n_K, (d')^n_L$. Namely, choose $(\alpha')^n : (K')^n \to (L')^n$ lifting the components $\alpha^n : K^n \to L^n$. As before we get a factorization

$$(K')^n \to K^n \xrightarrow{h^n} L^{n+1} \otimes_R I = I(L')^{n+1} \to (L')^{n+2}$$

of $(d')^n_L \circ (\alpha')^n - (\alpha')^{n+1} \circ (d')^n_K$. Then it is an pleasant calculation to show that

$$\omega^n_L \circ \alpha^n = (d^{n+1}_L \otimes \mathrm{id}_I) \circ h^n + h^{n+1} \circ d^n_K + (\alpha^{n+2} \otimes \mathrm{id}_I) \circ \omega^n_K$$

This proves the commutativity of the diagram in (2) of the lemma in this particular case. Using this for two different choices of bounded above free complexes representing $K$, we find that $\omega(K)$ is well defined! And of course (2) holds in general as well.

If $K$ lifts to $K'$ in $D^-(R')$, then we can represent $K'$ by a bounded above complex of free $R'$-modules and we see immediately that $\omega(K) = 0$. Conversely, going back to our choices $K^\bullet$, $(K')^n$, $(d')^n_K$, if $\omega(K) = 0$, then we can find $g^n : K^n \to K^{n+1} \otimes_R I$ with

$$\omega^n = (d^{n+1}_K \otimes \mathrm{id}_I) \circ g^n + g^{n+1} \circ d^n_K$$

This means that with differentials $(d')^n_K - g^n : (K')^n \to (K')^{n+1}$ we obtain a complex of free $R'$-modules lifting $K^\bullet$. This proves (1).

Finally, part (3) means the following: Let $R' \to S'$ be a map of rings. Set $S = S' \otimes_{R'} R$ and denote $J = IS' \subset S'$ the square zero kernel of $S' \to S$. Then given $K \in D^-(R)$ the statement is that we get a commutative diagram

$$\begin{array}{ccc} K \otimes^{\mathbf{L}}_R S & \xrightarrow[\omega(K)\otimes \mathrm{id}]{} & (K \otimes^{\mathbf{L}}_R I[2]) \otimes^{\mathbf{L}}_R S \\ \downarrow & & \downarrow \\ K \otimes^{\mathbf{L}}_R S & \xrightarrow{\omega(K \otimes^{\mathbf{L}}_R S)} & (K \otimes^{\mathbf{L}}_R S) \otimes^{\mathbf{L}}_S J[2] \end{array}$$

Here the right vertical arrow comes from

$$(K \otimes^{\mathbf{L}}_R I[2]) \otimes^{\mathbf{L}}_R S = (K \otimes^{\mathbf{L}}_R S) \otimes^{\mathbf{L}}_S (I \otimes^{\mathbf{L}}_R S)[2] \longrightarrow (K \otimes^{\mathbf{L}}_R S) \otimes^{\mathbf{L}}_S J[2]$$

Choose $K^\bullet$, $(K')^n$, and $(d')^n_K$ as above. Then we can use $K^\bullet \otimes_R S$, $(K')^n \otimes_{R'} S'$, and $(d')^n_K \otimes \mathrm{id}_{S'}$ for the construction of $\omega(K \otimes^{\mathbf{L}}_R S)$. With these choices commutativity is immediately verified on the level of maps of complexes. $\square$